

VALIDATION OF THE PROCEDURE OF IMPLEMENTATION OF LEAN STARTUP APPROACH (LSA) IN SME

Author(s)*: Isabella JESEMANN¹, Florian SELLNER², Sorin POPESCU³

Position: PhD Student¹, Student², Prof., PhD³

University: Technical University of Cluj-Napoca

Address: Cluj-Napoca, Memorandumului Str., No. 28, Romania

Email: Isabella.Jesemann@iao.fraunhofer.de¹, Florian.Sellner@iao.fraunhofer.de², sorin.popescu@muri.utcluj.ro³

Webpage: <http://www.utcluj.ro/>

Abstract

Purpose – Validating a proposed process to implement a new innovation method into small and medium-sized enterprises (SMEs) and deducting further steps in making the approach applicable. Helping SMEs to gain momentum in a globalized business environment.

Methodology/approach – In-depth interviews with experts from large enterprises, using the Lean Startup Approach (LSA), validation surveys with experts from both business and science.

Findings – The approach is useful and opens-up a valuable perspective for SMEs to innovate quickly and cost efficiently. Some aspects need to be more detailed and empirically validated.

Research limitations/implications – Since the approach has its roots in high-tech startup, a differentiated view on industry branches particular on SMEs would be important.

Practical implications – Further research and especially a pilot study inside SMEs would be beneficial.

Originality/value – Applying an innovation method that has so far mainly been used in startup and large enterprises by tapping into know-how from actual practitioners.

Key words: Innovation, SME, Globalization

Problem Statement and Overall Objective

Global megatrends like climate change, digital transformation, social inclusion movements and a further increasing globalization are affecting markets and businesses like never before (Jesemann et al., 2020: 595; Schaper, 2002: 525; Stocchetti et al., 2000: 3).

Digital change is the key challenge for industries as simple technical work will be replaced by machines and staff will have to meet new quality requirements. New business models based on disruptive innovation will be required.

This development has recently been reinforced by unexpected inflation, war in Ukraine and the COVID-19 pandemic (Zutshi et al. 2021), disrupting supply chains and driving up prices for producers and consumers. Consequently, governments release billions of aids and relief for enterprises and consumers all over the world, impose sanctions on warring parties and cut trade relations in favour of new partners, influencing supply and demand on a global scale.

All this makes the future more imponderable and thus poses greater risks for entrepreneurs making business decisions. Forecasts are becoming increasingly unreliable, making flexibility, agility and risk management of crucial importance. This applies to the area of product innovation and innovation management, as successful innovation often determines the medium and long-term business perspective of enterprises. Slow innovation without constant feedback from existing and potential customers exposes corporations to risks of being overtaken by their competitors or new global trends and thus being squeezed out of the market. Simultaneously, lost investments in newly developed products which fail on the market can be devastating, especially for smaller businesses. While startups are generally more agile due to their lean structure and their focus on new products and large enterprises hold many resources and dominant market positions, small and medium-sized enterprises (SMEs) are

often particularly vulnerable to such challenges (Gamage et al., 2020: 2), nevertheless they play an very important role in the economy of many countries and regions.

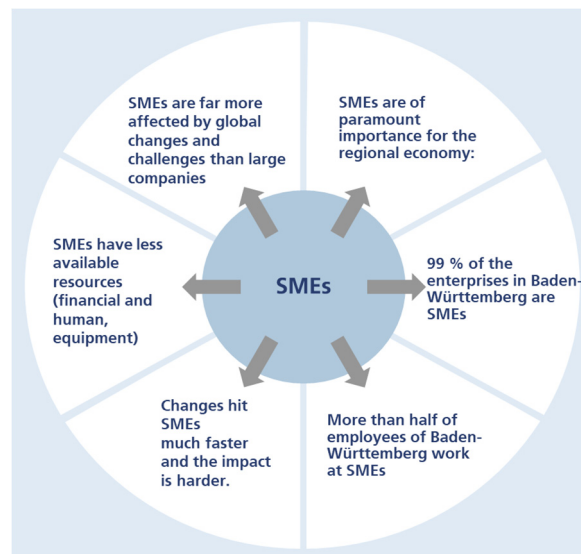


Figure 1: Role of SMEs in the economy and reasons for their vulnerability

Companies can only survive if they continuously innovate. However, this is not about improving existing products and processes. To be viable in the future and even to be able to play in the top league, incremental innovation is not enough. The survival strategy is to innovate disruptively, to react quickly and efficiently to market needs and to offer new products, solutions and services that meet the market needs. Thus, it becomes increasingly clear that traditional innovation methods struggle to keep up with the modern and fast changing needs of markets.

Prior Investigation

From these preliminary considerations, three main questions were formulated:

1. Which innovation methods are currently being used?
2. Which innovation method is most suitable and beneficial for SMEs?
3. How can it be implemented?

In first step, the authors conducted literature research in which the authors investigated traditional and current innovation methods, used by startups and large enterprises and compared them with each other.

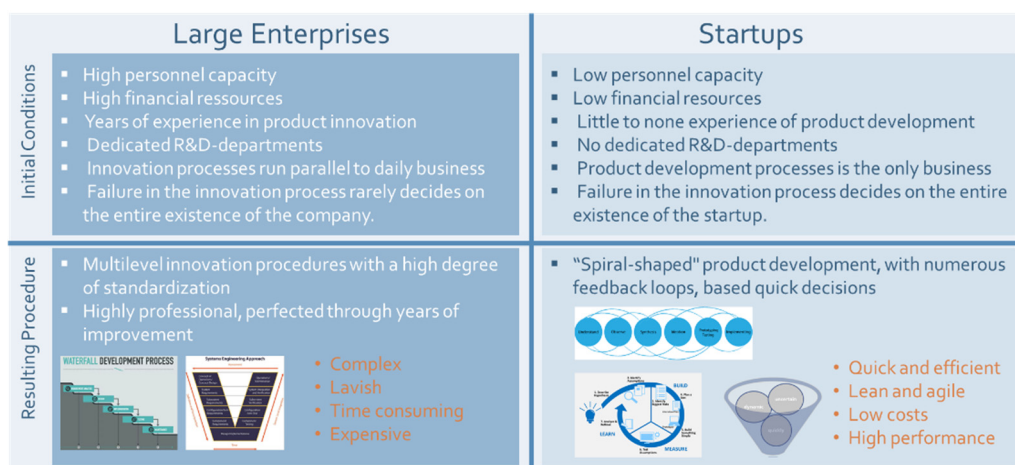


Figure 2: Conventional Innovation Methods in Large Enterprises vs Startups

The authors collected and evaluated the most promising strategies for innovation and weighed their strengths and weaknesses. They found the LSA to be the simplest and fastest and therefore most efficient way to innovate.

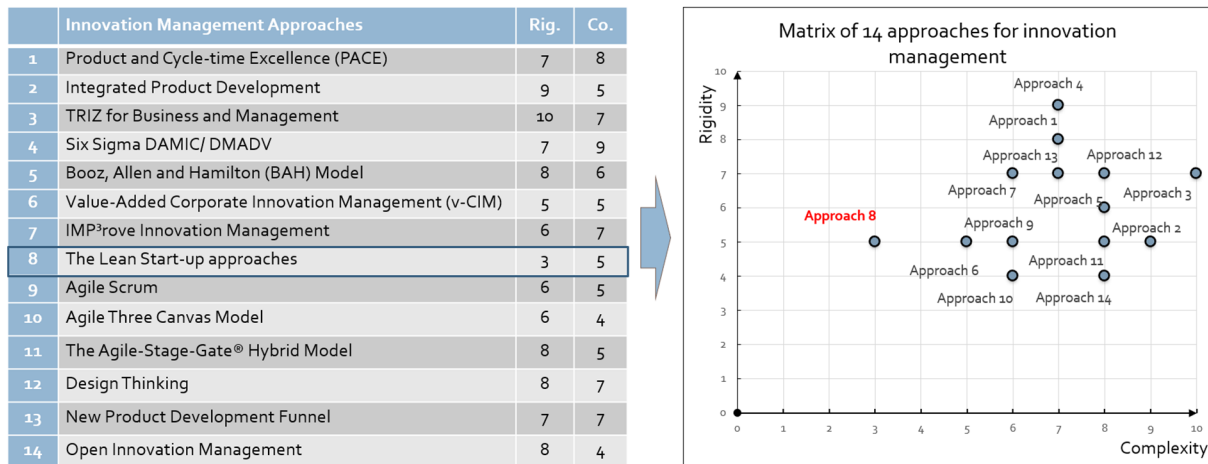


Figure 3: Comparing innovation management methods in terms of complexity and rigidity

Modern agile methods attempt to meet these challenges through a quicker, customer-oriented and iterative innovation process. One such approach is the Lean Startup Approach (LSA), a method used by innovative and fast-growing tech startups to innovate cost-effectively.

A central point of the method are customer interviews. The central point of product development is based on presenting a simple solution design called “Minimum Viable Product” (MVP) to the customer(s) and improving and developing the initial design to market readiness through permanent customer feedback.

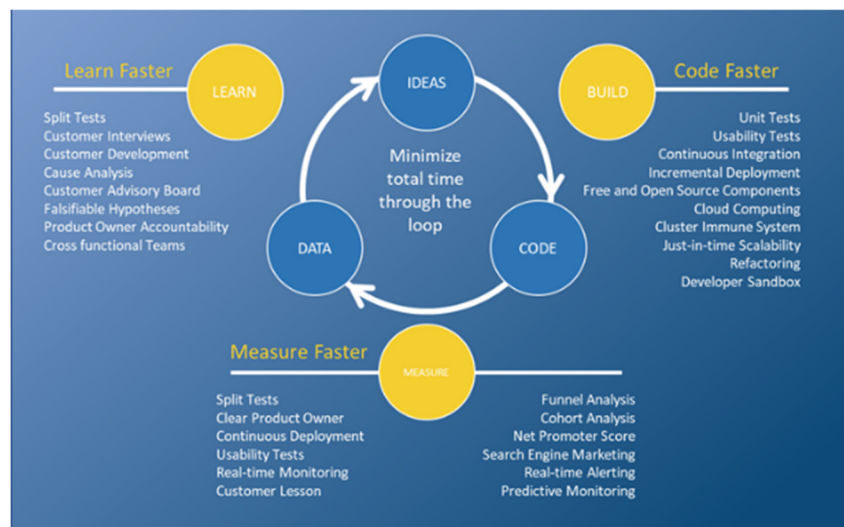


Figure 4: Graphic representation of the LSA according to Eric Ries

Through this customer-orientation, initial market approaches carry fewer risks of failing, while at the same time the process is more cost-efficient due to the use of MVPs and the innovation process able to be stopped or shifted at any time.

In next step the authors conducted expert interviews with practitioners from large enterprises already using the LSA to harness experiences in businesses greatly larger than startup. Challenges encountered by these, and other enterprises were then adopted as questions in the expert interviews.

The expert interviews conducted comprised of 15 experts from large enterprises and had experience with innovation management, particularly with the LSA. The main questions were:

- Why did the enterprise decide to try out LSA?
- What were the biggest challenges? What were the biggest benefits?

Although the experts came from different industries, countries and sectors, the responses were by and large very homogeneous, and the survey can be summarized as follows:

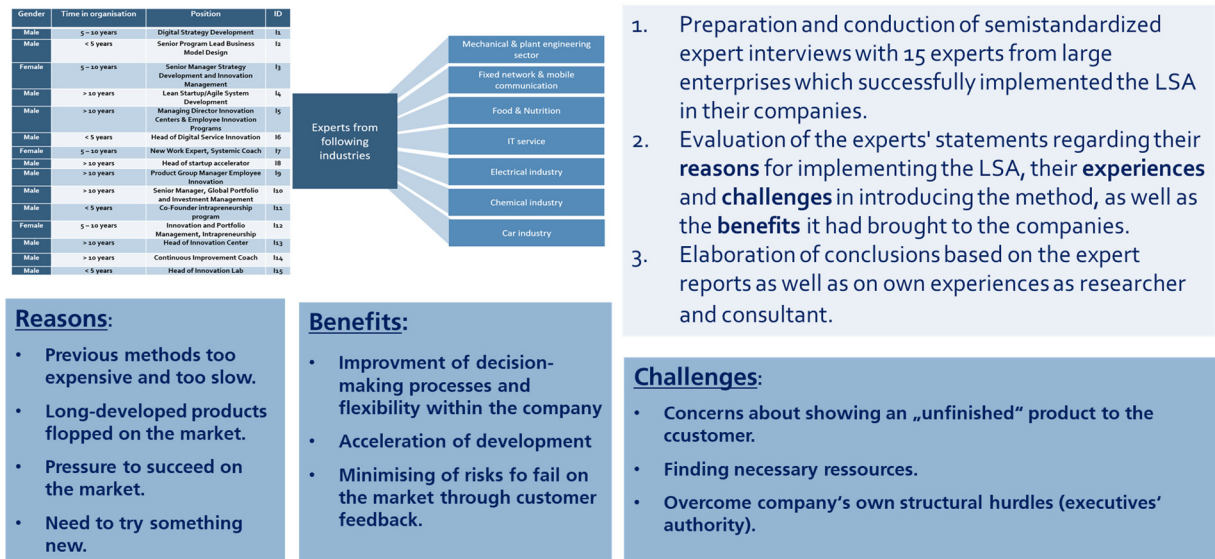


Figure 4: Interviews with Experts from Large Enterprises which implemented LSA

The experts gave prolific insights into the reasons why LSA was initially invented in their companies, their biggest challenges while implementing and their greatest benefits. They also described the difficult process of involving customers showing them unfinished MVPs yet collecting valuable feedback. The most important finding, however, was that the LSA can be migrated to and successfully applied in SMEs. The authors used these learnings to:

- describe a procedure for how to implement LSA in SMEs,
- give valuable preliminary advice for SMEs aiming to implement the LSA into their innovation framework,
- formulated step-by-step instructions with challenges that can be expected and possible solutions to these challenges.

The process was described in two sections:

1. Prerequisites and Preparations prior to the LSA implementation
2. Steps of the actual LSA-Implementation

The first section (*Prerequisites and Preparations*) consists of four issues or action areas that managements must address before the actual process of LSA implementation can begin. This section is divided into:

1. Preliminary considerations
2. Preliminary Actions
3. Upfront measures and
4. Familiarisation with the method.

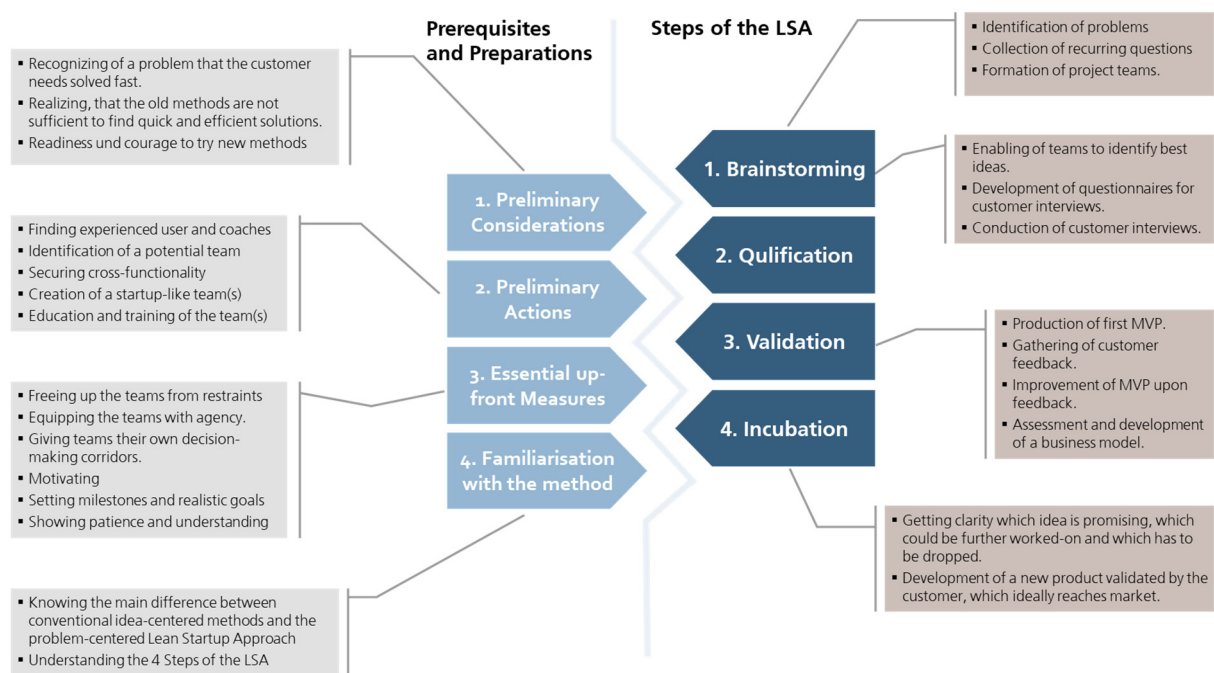


Figure 5: Process of implementation of the LSA in SMEs including Prerequisites and Preparations

The key point in this section is the realization that the company does not have a product that can survive on the market in the long term and that incremental innovation can no longer bring success. Each topic is complemented by possible challenges and valuable solutions based on the experience of the authors and the insights gained from the interviews with the experts.

The second section is a step-by-step guide consisting of the four steps for the implementation of the LSA. At each step, goals are defined, and exemplary time frames based on the experts are suggested.

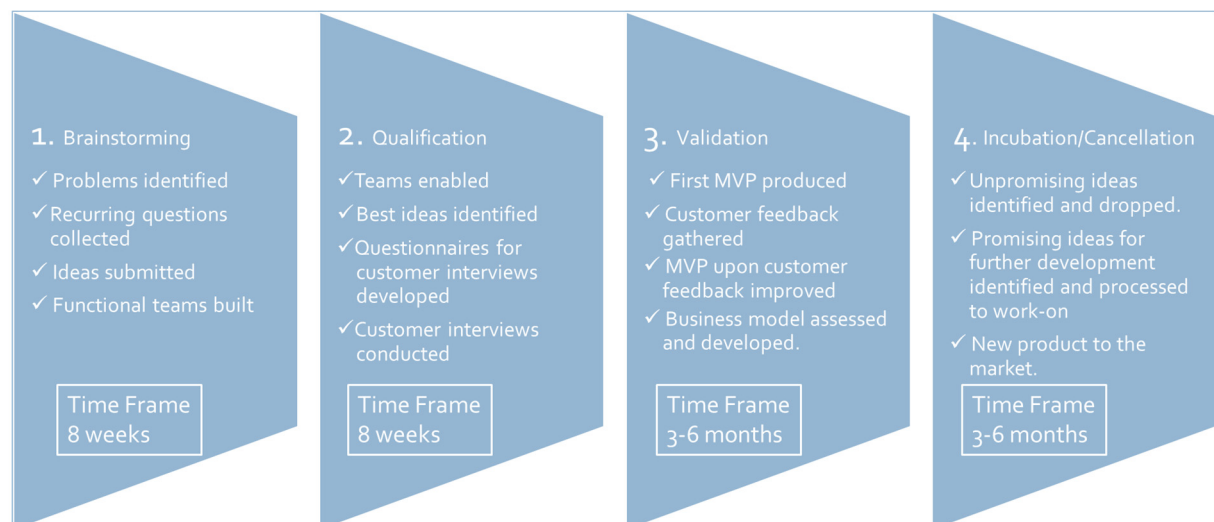


Figure 6: Four proposed steps to implement the LSA including goals and time frames

Scope of this Article

In this article, the authors aim to prove that the described approach to implementing LSA in SMEs is understandable, comprehensive, and feasible. The goal is to validate the defined step-by-step procedure for introducing the LSA in SMEs, the additional advice given on prerequisites and preparations, and the challenges, solutions and suggested good practices described. For this purpose,

13 international validators (1 Polish, 1 French/Tunisian, 1 German/Rumanian, 3 Romanian, 7 German, 7 male, 6 female) from business/SMEs (6), academia (2) and science/consulting (5), were selected to verify the above points and confirm their accuracy and completeness.

In addition to an extensive presentation on the subject area, the research goals and the description of the procedure, the validators received a questionnaire with questions regarding the completeness, comprehensibility, and feasibility of the procedure. Furthermore, they were asked to add comments and an overall assessment.

Questions regarding the Validator	- Name - Profession/Degree - Affiliation - Current area of responsibility - Experience with innovation management
Questions regarding the Prerequisites and Preparations	1. Is the process of implementation of the LSA in SMEs consisting of the <i>Preconditions and Preparations</i> and the <i>Step-by-step procedure</i> understandable and comprehensible? 2. Are the separate sections of the <i>Preconditions and Preparations</i> comprehensible? 3. Are the challenges described in the <i>Preconditions and Preparations</i> sufficient?
Questions regarding the LSA Procedure	4. Is the method description of the LSA in the <i>Preconditions and Preparations</i> understandable and sufficient? 5. Are the single steps of the LSA in the <i>Step-by-step procedure</i> understandable, comprehensible and complete? 6. Are the actions described in the <i>Step-by-step procedure</i> in each and every step comprehensible and complete? 7. Are the challenges described in the <i>Step-by-step procedure</i> in each and every step comprehensible and complete? 8. Are the solutions given in the <i>Step-by-step procedure</i> in each and every step comprehensible and sufficient? 9. Are the suggested good practices in the <i>Step-by-step procedure</i> in each and every step relevant and useful? 10. Are the timeframes in the <i>Step-by-step procedure</i> adequate?
Overall assesment	Possibility for the validator to add an overall statement or additional thought.

Figure 7: Questionnaire for the validation process

Results of the Validation

The overall feedback was positive. All validators saw a need to support SMEs at their innovation management. Validators 2, 10 and 11 answered yes to every question. Validator 10 additionally confirmed the "yes" with a comment.

On question 1 validator 1 commented that it might be difficult for companies to realize that the old method is no longer sufficient. Validator 3 criticised that speed is not the only topic SMEs should focus on while innovating. In contrast, according to expert 7, SMEs tend to ignore the fact that they need a new approach for as long as possible, so when they notice they need a quick solution. This underlines the authors' thesis regarding importance of speed.

On question 2 validators 10 and 13 remarked that "all [preconditions] merge and influence each other] and therefore should not be seen separately.

On question 3 validator 1 suggested adding industry-specific challenges, while validator 4 demands inclusion of innovation culture of the company. Validators 10 and 13 found the described actions understandable, but being no dedicated LSA experts, could not verify completeness.

On question 5 validator 3 wished more details and expert 4 felt that "there should be a Step 5 after incubation where the solutions are transferred to regular activity and the cycle is restarted."

There were no additional comments on **question 4**.

On question 5 validators 3 and 7 wished more details regarding practical use, while validator 4 suggested implementing a step 5 after incubation in which "the solutions are transferred o regular activity and the cycle is restarted"

On question 6 validator 1 noted that more details should be given on how to gather information about customers. Meanwhile validator 3 missed "a quality gate" after each step.

On question 7 validator 1 proposed adding “industry-specific challenges” and validator 6 suggested integrating two different types of interviews 1) customer discovery, where unique “pain points” are pointed out, before moving on to 2) validating MVPs.

On question 8 validators 3, 4 and 7 sought for more details. Validator 9 saw a challenge in the existing innovation culture.

On question 9 validator 4 stated that the solutions and good practices given are a good starting point but suggested they could be expanded and detailed as well as supplemented with examples. Validators 3 and 7 suggested a more detailed description, too.

On question 10 validator 6 remarked that the time frames given might be object to very high variability and need to be backed up by data.

In the **overall assessment** section, almost all validators took the opportunity to add comments. Validators 1 and 2 considered the approach very interesting and valuable. Validator 2 said: “It can bring great value as supplement to the existing innovation methods”. Validator 11 stated the process description was very well presented and regards “the approach [as] understandable and in line with practical experience”. While validator 8 wished for an extended description for SMEs, validator 12 suggested a “fast-track procedure” for SMEs.

Validator 9 was remarkable sceptical, seeing the main contribution of the authors only in the description of the preconditions. He also questioned whether LSA that originates from the technology startups would also work in manufacturing companies without adaptations. Further, he noted that the success of startups lies in their ability to create new business models and educate the market on how to adapt to technological disruption. In this regard, LSA would have shortcomings as innovators listen only to individual customers rather than creating new markets for new products. In his opinion the unique value propositions combined with a new business model is key to success rather than “too much feedback from individual customers”. In contrast, the other 12 validators agreed on the great potentials and advantages of the method.

Overall, a slight disparity between respondents from business, academia and consulting could be observed. While the proposed steps were mostly very well received, validators from academia/consulting were particularly positive. This can best be demonstrated with a quote from validator 2: “For me as a researcher and consultant on innovation everything is evident and comprehensible. The approach is very interesting and modern. It can bring great value as supplement to the existing innovation methods. However, from my experience I would expect SMEs to be sceptic out of fear that not every possible challenge has been addressed and explained and something could be missing. It is therefore important that the process description not only recommends the use of a coach but makes this a prerequisite”.

Conclusion and Implications for Future Research

Besides the predominantly positive feedback, the validators contributed some aspects to complement the research. Some comments like investigation and consideration of varying innovation cultures of enterprises as a factor of influence of innovation activity were considered in the authors’ research but were not included in the briefing.

One suggestion mentioned several times was the inclusion of specific requirements, peculiarities and challenges of companies operating in different industries, especially with a differentiated analysis of technology or production companies. However, based on the results of the expert survey conducted prior to the validation, the authors believe that the method is applicable without predefined industry specifications since the interviewed experts came from a wide variety of industries. The adaptation to the needs of individual branches or even companies should therefore be worked out individually in the preparation phase in cooperation with the coach, because here the specification can be tailored exactly to the needs of the company. A predefined industry specification would elevate the method to a meta-level and restrict it.

The question of the applicability of LSA in SMEs was also raised. The expert interviews provided the statement that the method can also be applied in SMEs, and no indications were discernible that the approach would have to be fundamentally modified. On the contrary, the authors assume that all

necessary adjustments can also be made in the preparatory phase. A specification that is too concrete would go far beyond the generally applicable process description.

The question regarding the actual purpose of the customer interviews, namely are they used to evolve the product only or the overlaying business model, seems justified to the authors. Therefore a two-step-interviews including both topics will be considered in further research.

In summary, the two most important findings of the validation upon all experts agreed were that

1. there is a great need to support SMEs in their innovation activities and
2. that LSA is a valuable supplement to the existing innovation methods.

Interesting finding was the difference in the comprehensibility of the described process between the validators from academia, business, and science/consultancy. While the academic validators were focused mainly on the formalities, and the SME representatives feared gaps in the description of the process and wanted more details or an accompaniment by a coach, the researcher/consultants seemed to be best able to deal with the information given. It is explained by their scientific research and practical approach to innovation management. After all, they would be the coaches in such processes.

It became clear, that to be applicable especially by smaller enterprises, the LSA should be coached and facilitated from outside of the company. This is consistent with the findings from the expert interviews. Since the approach is loosely formalized by its nature, it thrives on practical experience, which raises the need of a "hands-on" mentality and outside know-how. The expert interviews stressed that most of the concerns (like presentation of an MVP or how to conduct customer interviews) dissolved quickly once the first hurdle of trying it has been overcome. Hence, the most important step is the decision to open-up to the approach. Since every company is different, each company does benefit from figuring out the specifics of each step in their own business exploratively, regarding their own products, processes, and customers.

Overall, LSA may seem intimidating at first glance, as it is rather vague and not very formalized, and the customer interviews seem unfamiliar. However, therein lies its greatest strength. Few formalities give maximum flexibility and constant exchanges with customers provide immediate validation. This leads to the fast pace and low resource requirements, making LSA an effective method for companies of all sizes. However, to realize the true potential of the approach, the involvement of external coaches is the key factor.

References

- Gamage, S. K. N., E. M. S. Ekanayake, G. A. K. N. J. Abeyrathne, R. I. P. R. Prasanna, J. M. S. B. Jayasundara, P. S. K. Rajapakshe**
2020
"A Review of Global Challenges and Survival Strategies of Small and Medium Enterprises (SMEs)" In *Economies* 2020, Vol. 8, 79, MDPI.
- Jesemann, I., T. Beichter, K. Herburger, C. Constantinescu and M. Rüger**
2020
"Migration of the Lean-Startup approach from High-Tech startups towards product design in large manufacturing enterprises." In *Procedia CIRP* 18 August 2020 Vol. 91: 594-599.
- Schaper, M.**
2002
"The challenge of environmental responsibility and sustainable development: Implications for SME and entrepreneurship academics." In *Radical changes in the world: Will SMEs soar or crash:* 541-553.
- Stocchetti, A., A. Mundim, A. Rossi**
2000
"SMEs in Global Market: Challenges, Opportunities and Threats" In *Brazilian Electronic Journal of Economics*, 2000, Vol. 3, 1.
- Zutshi, A., J. Mendy, G. D. Sharma, A. Thomas, T. Sarker**
2021
"From Challenges to Creativity: Enhancing SMEs' Resilience in the Context of COVID-19" In *Sustainability* Vol. 13, 12: 6542.